Technological University Dublin - Blanchardstown

Academic Term	2019-20
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Programme Code	Programme Title	Module Code
BN002	Higher Certificate in Science in Computing in Information Technology	COMP H2026
BN013	Bachelor of Science in Computing in Information Technology	COMP H2026
BN034	Bachelor of Science in Computing in Digital Forensics & Cyber Security	COMP H2026
BN104	Bachelor of Science (Honours) in Computing	COMP H2026
BN120	Bachelor of Science (Honours) in Computing in Digital Forensics & Cyber Security	COMP H2026

Module Title	Information Technology Mathematics

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Dr. Simon Caton

Instructions to candidates:

1) **To ensure that you take the correct examination, please check that the module and programme which you are following is listed in the table above.**
2) **Question 1 is compulsory. Candidates should attempt ALL parts of Question 1 and any other TWO questions.**
3) **This paper is worth 100 marks. Question 1 is worth 40 marks. The remaining questions are worth 30 marks each.This is a 2 hour exam.**
4) **Remember to mark each number and section clearly and to list the questions which you have attempted in the correct order on the front of your exam answer script.**
5) **Show all your work.**

DO NOT TURN OVER THIS PAGE UNTIL YOU ARE TOLD TO DO SO

Question 1: (Compulsory)

Attempt ALL Parts

a) Given the following matrices: $\boldsymbol{A} = \begin{bmatrix} \mathbf{1} & \mathbf{-3} \\ \mathbf{0} & \mathbf{2} \\ \mathbf{4} & \mathbf{-2} \end{bmatrix}$ $\boldsymbol{B} = \begin{bmatrix} \mathbf{2} & \mathbf{-2} & \mathbf{0} \\ \mathbf{3} & \mathbf{1} & \mathbf{1} \end{bmatrix}$

i. What is the size of each matrix *A* and *B*?
ii. Write down the value of the elements $\boldsymbol{a_{22}}$ and $\boldsymbol{b_{23}}$.
iii. Write down the transpose of matrix *B*.
iv. Find the matrix given by $\mathbf{2}(\boldsymbol{A} + \boldsymbol{B^T})$.

(5 marks)

b) Define the terms *Graphs* and *Trees*. Provide <u>one</u> example to outline the differences.

(5 marks)

c) Evaluate *t*(2) and *t*(3) and *t*(4) for the following recursively defined sequence:

t(1) = 3
t(n) = *t*(*n*-1) + 4 (*n*>1) **(5 marks)**

d) Explain the statistical terms *mode*, *mean* and *median*. Include in your answer the relationship between them.

(5 marks)

e) Using the data 2, 12, 17, 19, 25, 32 calculate the

i. Mean
ii. Standard Deviation **(5 marks)**

f) If two fair dice are thrown:

i. What is the probability of getting a three and a two?
ii. What is the probability that the PRODUCT of the numbers will be greater than or equal to three?

(5 marks)

g) When differentiating the function $\boldsymbol{f(x)} = \mathbf{3}\boldsymbol{x} + \mathbf{5}$ from first principles we obtain the equation: $\frac{\boldsymbol{dy}}{\boldsymbol{dx}} = \lim_{\boldsymbol{h \to 0}} \frac{\boldsymbol{\alpha x + \beta h + 5 - 3x - 5}}{\boldsymbol{h}}$. What are the values of $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$?

(5 marks)

h) Evaluate the following definite integral: $\int_1^3 (\mathbf{2}\boldsymbol{x^2} + \mathbf{3}\boldsymbol{x} + \mathbf{1}).\boldsymbol{dx}$ **(5 marks)**

<u>Total: 40 marks</u>

Question 2: Statistics and Probability

a) Describe the difference between:

i. Observational Data and Experimental Data
ii. Random Sampling and Stratified Random Sampling **(4 marks)**

b) For each of the scatter plots below in Figure 2.1, state whether there is correlation and if so, state the type and strength of that correlation.

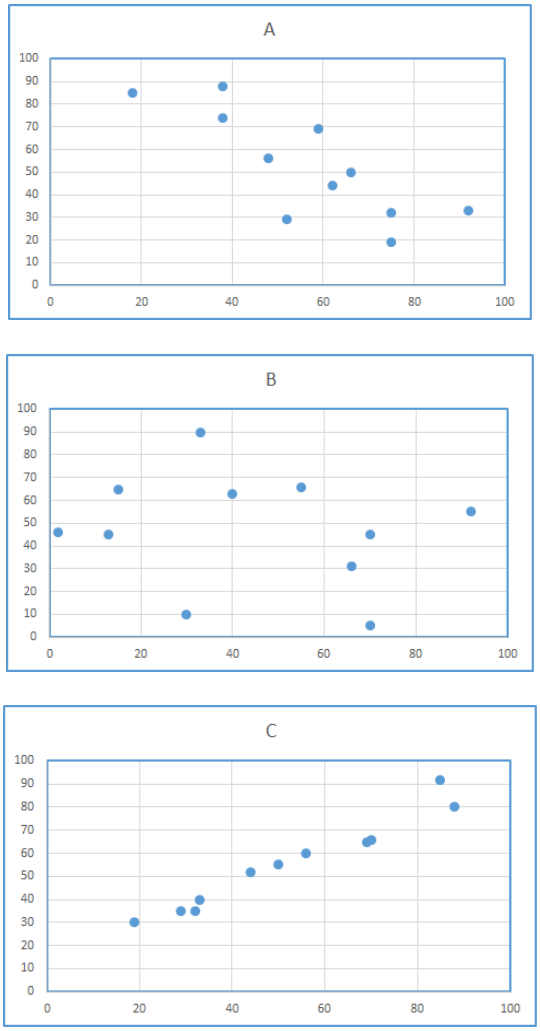


Figure 2.1

(3 marks)

Question is continued on the next page.

c) Using the following table (Table 2.1):

Table 2.1

Class Interval	Frequency
2 and < 4	3
4 and < 6	18
6 and < 8	9
8 and < 10	7

i. Compute the *Mean* of the grouped data. **(3 marks)**
ii. Calculate the *Standard Deviation* of the grouped data. **(5 marks)**
iii. Draw a *histogram* for the data. **(4 marks)**
iv. Comment on the *symmetry* or *skewness* of the distribution. **(2 marks)**

d) A local sports club holds a weekly lottery draw based around picking 4 correct numbers from 32. What is the probability of winning in this lotto draw with a single ticket? **(3 marks)**

e) 5% of the items produced by a machine in a factory are defective. 40 items are chosen at random. Using the *binomial distribution*, find the probability that 3 of the chosen items are defective. **(6 marks)**

Total: 30 marks

Question 3: Matrices and Recursion

a) Write down the *identity matrix* for a 2x2 and 3x3 matrix respectively. **(3 marks)**

b) Given the following matrices:

$$\boldsymbol{A} = \begin{bmatrix} \mathbf{4} & \mathbf{-2} \\ \mathbf{2} & \mathbf{-2} \end{bmatrix} \qquad \boldsymbol{B} = \begin{bmatrix} \mathbf{-1} & \mathbf{3} \\ \mathbf{1} & \mathbf{2} \end{bmatrix} \qquad \boldsymbol{C} = \begin{bmatrix} \mathbf{3} & \mathbf{-1} \end{bmatrix}$$

i. Calculate $\mathbf{2}\boldsymbol{C}(\boldsymbol{B} - \boldsymbol{A})$. **(4 marks)**

ii. Find the *determinant* and *inverse* of matrix $\boldsymbol{A}$. **(5 marks)**

c) Calculate the inverse of the 3x3 matrix:

$$\boldsymbol{D} = \begin{bmatrix} \mathbf{-1} & \mathbf{3} & \mathbf{6} \\ \mathbf{1} & \mathbf{2} & \mathbf{0} \\ \mathbf{0} & \mathbf{3} & \mathbf{2} \end{bmatrix}$$

(9 marks)

d) Give a definition of *recursion* and explain why it is an important concept in computer science. **(4 marks)**

e) Find a *recursive* and a *non-recursive* definition for the following sequence of positive even natural numbers: **(5 marks)**
2, 4, 6, 8, 10, 12, …

Total: 30 marks

Question 4: Graphs and Trees

a) What is a *Rooted Tree* structure? In your answer provide an example of a rooted tree representation to explain the following concepts: *root node*; *parent node*; *child node*; *leaf node*. **(5 marks)**

b) Draw the graph with the following adjacency matrix.

$$\begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 1 & 2 & 0 \end{pmatrix}$$

(4 marks)

c) Verify for the graph in part b) that the sum of the degrees of the vertices equals twice the number of edges. **(4 marks)**

d) The relative costs of building the links between 6 nodes denoted by $A, B, C, D, E \text{ and } F$ in a proposed local area network is shown below in table 4.1.

Table 4.1

	A	**B**	**C**	**D**	**E**
B	85				
C	-	95			
D	65	50	-		
E	55	-	-	70	
F	45	-	80	-	-

Using *Prim's Algorithm*, find the network with the least possible cost that could be built to link all the nodes in the network. **(10 marks)**

e) Describe, with the aid of examples, the difference between *breadth-first* and *depth-first* graph traversals. **(7 marks)**

Total: 30 marks

Formulae

Matrix Determinant

$\det A = ad - bc$ (2x2 matrix)

$\det A = a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13}$ (3x3 matrix using row 1)

Matrix Inverse

$A^{-1} = \frac{1}{\det A}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ (2x2 matrix)

$A^{-1} = \frac{1}{\det A}C^T$, where *C* is the matrix of Cofactors (3x3 matrix)

General Probability:

If two events A and B are mutually exclusive, then the following law holds:

$P[A \text{ or } B] = P[A] + P[B]$

If the two events are not mutually exclusive, the law above is broadened so that the probability of A or B occurring is given by:

$P[A \text{ or } B] = P[A] + P[B] - P[A \text{ and } B]$

Consider two events *A* and *B* with event *B* dependent on *A*. The probability of the occurrence of both events is given by:

$P[A \text{ and } B] = P[A] \times P[B \mid A]$

Probability Distributions:

$P[X = r] = {}^nC_r\, p^r q^{n-r} = \frac{n!}{(n-r)!r!}\, p^r q^{n-r}$ (Binomial distribution)

$P[X = r] = \frac{e^{-\mu}\mu^r}{r!}$, where $\mu = np$ (Poisson distribution)

$$a = \frac{x - \mu}{\sigma}$$ (Normal distribution)

Statistics:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$ (Dataset mean)

$$s = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}} \quad \text{or} \quad s^2 = \frac{\sum_{i=1}^{n} x_i^{\,2} - n\left(\bar{x}\right)^2}{n-1}$$ (Dataset standard deviation)

$$\bar{x} = \frac{\sum_{i=1}^{n} f_i m_i}{\sum_{i=1}^{n} f_i}$$ (Grouped data mean)

$$s = \sqrt{\frac{\sum_{i=1}^{n} f_i (m_i - \bar{x})^2}{\sum_{i=1}^{n} f_i - 1}} \quad \text{or} \quad s^2 = \frac{\sum_{i=1}^{M} f_i m_i^{\,2} - M\left(\bar{x}\right)^2}{M-1}$$ (Grouped standard deviation)

Derivation:

First Principles: $\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ **Power Rule:** $\frac{dy}{dx} = nx^{n-1}$

Integral: $\int \boldsymbol{ax^n dx} = \boldsymbol{\frac{ax^{n+1}}{n+1} + c}\,,\ \boldsymbol{n \neq -1}$

Definite Integral:

$$\int_a^b f'(x)dx = f(x)\Big]_a^b = f(b) - f(a)$$

Table of Probabilities – P[*Z* > a], where *Z* is the standard normal variable.

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002
3.6	0.0002	0.0002	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.7	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.8	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4.0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000